

The Cowan Atomic-Structure Codes

What RCN, RCN2, RCG, RCE Actually Compute, from First Principles

A working explainer for the Modern Atomic Linelists (`mal`) project

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Abstract

This document explains, from first principles, what the four Cowan programs do and how they chain together to produce atomic energy levels and oscillator strengths (*gf*-values) — the data that go into stellar line lists. It maps the physics onto the *actual* programs, input decks, and output files in this repository (`cowan_nist/extracted/`). The goal is a shared, durable understanding of the engine we are modernizing, with particular attention to the least-squares fitting stage (RCE), which is the heart of the project. The canonical reference throughout is R. D. Cowan, *The Theory of Atomic Structure and Spectra* (Univ. California Press, 1981), cited as **TASS**.

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1 The big picture: why these four programs exist

A stellar line list needs, for each transition, the lower and upper energy levels, the wavelength, and the oscillator strength *gf*. The wavelength is just the level difference; the levels and the *gf* are the hard part. Computing them *ab initio* for complex atoms (open *d*- and *f*-shells, the iron group) is not reliably accurate. Cowan’s insight, embodied in these codes, is a *semi-empirical* scheme: compute the atomic structure approximately from theory, then *adjust a small number of physically meaningful radial parameters until the predicted levels match the experimentally measured ones*. The wavefunctions (eigenvectors) that result are far better than the purely theoretical ones, and the *gf*-values computed from them are correspondingly better.

The pipeline in one breath: RCN solves for the radial wavefunctions and the *radial integrals* ($F^k, G^k, \zeta_{n\ell}, \dots$) of each electron configuration; RCN2 forms the configuration-interaction and

electric-multipole (transition) radial integrals between configurations; RCG builds the angular part, assembles and *diagonalizes* the energy (Hamiltonian) matrix to get energy levels and eigenvectors, and from the eigenvectors computes *gf*-values; RCE re-runs the diagonalization inside a *least-squares loop*, varying the radial parameters until the computed levels best fit the observed levels.

The division of labor reflects a deep structural fact about atomic Hamiltonians: the energy of a level separates into **radial** factors (integrals over the one-electron radial wavefunctions — these are the F^k , G^k , $\zeta_{n\ell}$, R^k) multiplied by **angular** factors (pure angular-momentum algebra, the same for any atom with a given configuration). RCN/RCN2 compute the radial side; RCG computes the angular side and combines them; RCE treats the radial numbers as adjustable.

2 The physics each stage implements

2.1 The central-field starting point and rcn

We seek the eigenstates of the many-electron Hamiltonian — the non-relativistic electronic Hamiltonian plus the leading (spin-orbit) relativistic correction,

$$H = \underbrace{\sum_i \left(-\frac{1}{2} \nabla_i^2 - \frac{Z}{r_i} \right) + \sum_{i < j} \frac{1}{r_{ij}}}_{\text{non-relativistic}} + \underbrace{\sum_i \xi(r_i) (\mathbf{l}_i \cdot \mathbf{s}_i)}_{\text{spin-orbit}}, \quad (1)$$

in atomic units. The *central-field approximation* writes each electron’s wavefunction as a product of a radial part $P_{n\ell}(r)/r$ and angular/spin parts. RCN solves the resulting one-electron radial equations self-consistently for a *single configuration* (e.g. $3d^7$, or $3d^6 4p$), using one of several approximations to Hartree–Fock; in practice the Hartree-plus-statistical-exchange (HX) method or center-of-gravity HF is used, optionally with relativistic corrections (the “HFR” / “HXR” options — important for the iron group and heavier).

A converged set of these single-electron radial functions is shown in Fig. 1: $P_{n\ell}(r) = r R_{n\ell}(r)$ is the quantity the code actually carries on its radial mesh, with $\int_0^\infty P_{n\ell}^2 dr = 1$. These are the *actual* self-consistent-field orbitals from our build of RCN for Sn^{7+} ($4d^7$), our worked example (§4). Note the nodal structure ($n - \ell - 1$ nodes: $1s$ nodeless, $2s$ one node, ...) and that the orbitals are pulled in to small radius by the high nuclear charge ($Z = 50$): the $1s$ peaks near $r \approx 0.02 a_0$, while the $4d$ valence orbital peaks near $0.9 a_0$.

From the converged radial wavefunctions $P_{n\ell}(r)$, RCN computes the **radial integrals** that parametrize the energy:

$$F^k(n\ell, n'\ell') = \int \int P_{n\ell}^2(r_1) \frac{r_{<}^k}{r_{>}^{k+1}} P_{n'\ell'}^2(r_2) dr_1 dr_2 \quad (\text{direct Slater integral}), \quad (2)$$

$$G^k(n\ell, n'\ell') = \int \int P_{n\ell}(r_1) P_{n'\ell'}(r_1) \frac{r_{<}^k}{r_{>}^{k+1}} P_{n\ell}(r_2) P_{n'\ell'}(r_2) dr_1 dr_2 \quad (\text{exchange Slater integral}), \quad (3)$$

$$\zeta_{n\ell} = \frac{\alpha^2}{2} \int P_{n\ell}^2(r) \frac{1}{r} \frac{dV}{dr} dr \quad (\text{spin-orbit parameter}), \quad (4)$$

plus E_{av} , the configuration’s center-of-gravity energy. The $r_{<}^k/r_{>}^{k+1}$ factor is the multipole expansion of the electron–electron repulsion $1/r_{12}$; Fig. 2 shows this kernel and the integrand it produces. These few numbers per configuration are the *radial parameters*. Everything semi-empirical happens by treating them as adjustable later (§2.4).

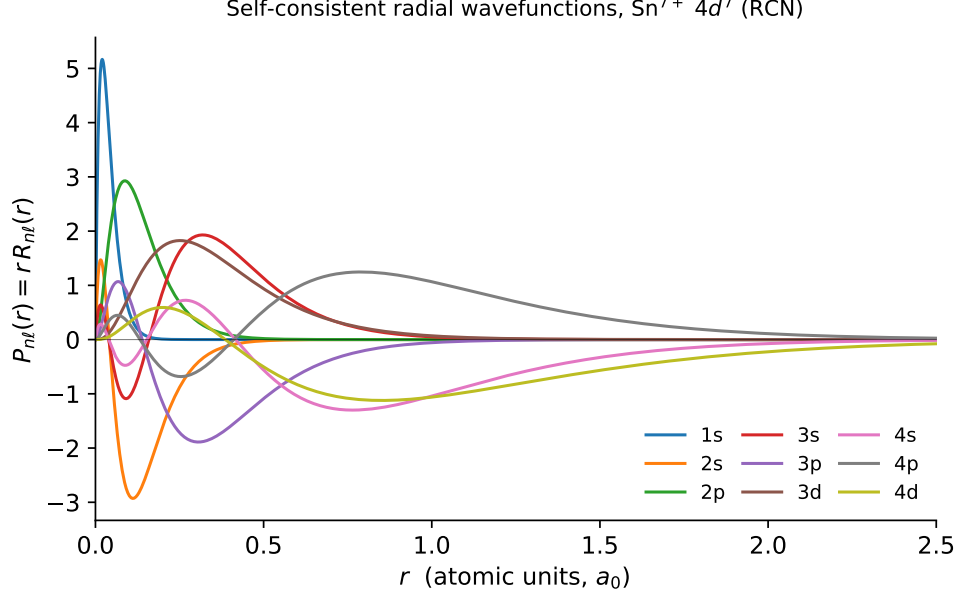


Figure 1: Self-consistent reduced radial wavefunctions $P_{n\ell}(r) = r R_{n\ell}(r)$ for $\text{Sn}^{7+} 4d^7$, computed by our build of RCN (the nine occupied subshells 1s through 4d). Each is normalized to $\int P_{n\ell}^2 dr = 1$. This is real output of the engine, not a model.

2.2 Configuration interaction and transition integrals: rcn2

Real atomic states are not pure single configurations; nearby configurations of the same parity and J *mix* (configuration interaction, CI). RCN2 takes the radial wavefunctions from RCN for *several* configurations and computes:

- the **CI Slater integrals** R^k (the r_{ij}^{-1} matrix elements *between* different configurations), which set the strength of the mixing (it also computes configuration overlaps and the off-diagonal spin-orbit integrals);
- the **electric-multipole radial integrals** $\langle n\ell | r | n'\ell' \rangle$ (dipole, and higher multipoles), the radial part of the transition moment — the seed of every *gf*-value.

2.3 Angular algebra, diagonalization, and where *gf* comes from: rcg

RCG supplies the *angular* half. Using angular-momentum (Racah/Wigner) algebra and tabulated coefficients of fractional parentage (cfp), it builds the Hamiltonian matrix in a chosen coupling basis. Schematically, every matrix element of (1) is

$$\langle \gamma' | H | \gamma \rangle = \sum_{\text{integrals}} \underbrace{c_{\gamma'\gamma}^{(\cdot)}}_{\text{angular coeff. (RCG)}} \times \underbrace{\{E_{av}, F^k, G^k, \zeta_{n\ell}, R^k\}}_{\text{radial integrals (RCN/RCN2)}}. \quad (5)$$

RCG then **diagonalizes** this matrix for each J . The eigenvalues are the predicted **energy levels**; the eigenvectors give the **composition** of each level (its percentage breakdown over basis states — e.g. “72% 5D + 18% 3F + ...”).

The oscillator strength follows from the eigenvectors. The line strength between levels βJ and $\beta' J'$ is

$$S = \left| \langle \beta J \| \mathbf{P}^{(1)} \| \beta' J' \rangle \right|^2, \quad gf = \frac{8\pi^2 m_e c}{3h e^2} \sigma S \propto \sigma S, \quad (6)$$

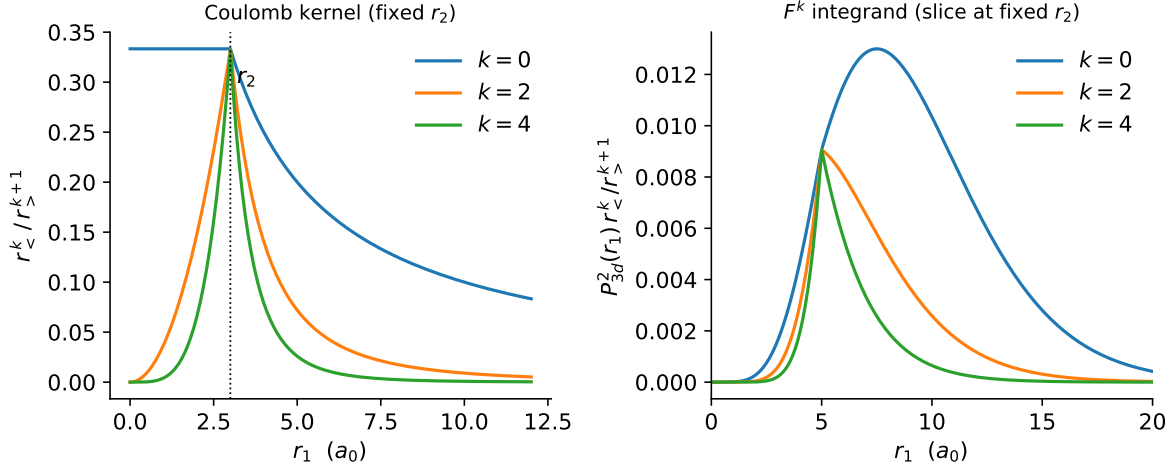


Figure 2: *Left*: the Coulomb multipole kernel $r_{<}^k/r_{>}^{k+1}$ versus r_1 at fixed r_2 — continuous but with a cusp at $r_1 = r_2$, and increasingly short-ranged (peaked near $r_1 = r_2$) for higher k . *Right*: the resulting F^k integrand $P_{3d}^2(r_1) r_{<}^k/r_{>}^{k+1}$ (slice at fixed r_2): the higher- k multipoles weight the region where the two electrons are close, so $F^0 > F^2 > F^4$. (*Hydrogenic 3d, illustrative.*)

where σ is the transition wavenumber and the reduced matrix element of the dipole operator $\mathbf{P}^{(1)}$ is a sum over the RCN2 transition radial integrals weighted by *both* levels' eigenvector components. Here gf is the *weighted* oscillator strength (g already folded in — there is no division by g ; $f = gf/g$ with g the statistical weight of the lower level). In practice RCG evaluates $gf = 3.0375 \times 10^{-6} \sigma S$ with σ in cm^{-1} and S in atomic units ($e^2 a_0^2$). This is the crucial coupling that drives the whole project: **many lines' gf -values depend on the same few radial parameters through the shared eigenvectors.** A small change in the fit redistributes oscillator strength among coupled lines.

2.4 The semi-empirical least-squares fit: rce

This is the stage we are modernizing, so we treat it carefully. RCG's predicted levels are only as good as the theoretical radial integrals — and those are systematically off: Hartree–Fock *overestimates* the Slater integrals by ~ 10 – 20% (so the computed term splittings come out too large; the customary remedy is to scale them down by ~ 0.85). RCE fixes this empirically. Its job (TASS §16-3 to 16-5):

1. Treat the radial parameters $\mathbf{x} = \{E_{\text{av}}, F^k, G^k, \zeta_{nl}, R^k, \dots\}$ as *free*. The angular coefficients in (5) are fixed (precomputed by RCG and passed in the binary file **TAPE2E**).
2. For trial $\mathbf{x}$, build and diagonalize the energy matrix for each J to get computed levels $E_i^{\text{calc}}(\mathbf{x})$.
3. Minimize the least-squares residual against the *experimentally observed* levels E_i^{obs} (from NIST/lab),

$$\chi^2(\mathbf{x}) = \sum_{i \in \text{included}} \left[E_i^{\text{calc}}(\mathbf{x}) - E_i^{\text{obs}} \right]^2, \quad (7)$$

iterating until the parameters stop changing (by < 0.03 in their units between cycles) or a cycle cap is hit. In standard RCE each level is simply *included* or *excluded* (no per-level weights); parameters can also be held fixed, or tied in fixed ratios (a group forced to scale together). Adding per-level or per-line weighting is one of the modernizations we plan, not existing behavior.

4. Feed the *fitted* $\mathbf{x}$ back through the diagonalization to get the *improved* eigenvectors — and hence the improved *gf*-values (via RCG).

The effect of the fit is sketched in Fig. 3: the raw *ab initio* term pattern is too spread out, and adjusting the radial parameters slides the computed levels onto the observed ones — simultaneously improving every eigenvector, and therefore every *gf*.

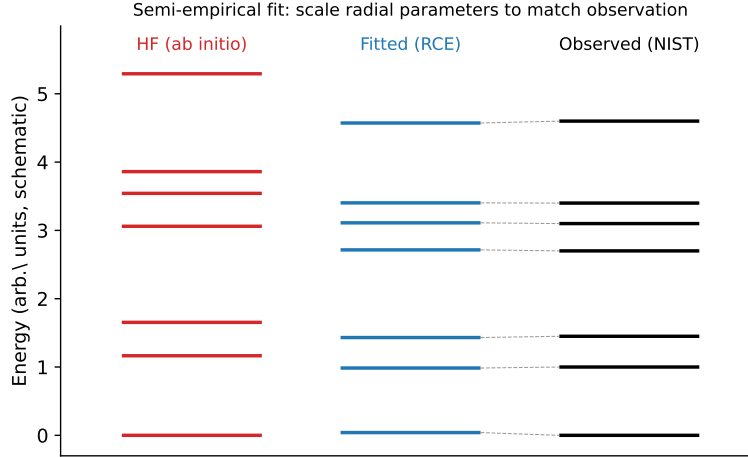


Figure 3: Schematic of the semi-empirical fit. Raw Hartree–Fock levels (left) are systematically too widely split because the Slater integrals are $\sim 15\%$ too large; the RCE least-squares adjustment of the radial parameters (middle) brings the computed levels into agreement with the observed NIST levels (right; dashed guides connect matched levels). (*Schematic, illustrative pattern.*)

The fit in (7) is ill-conditioned: parameters are correlated, the problem is non-linear, and which configurations to include and which parameters to free/fix/tie is a matter of hand-tuned judgment with “no strict methodology.” This is the only non-automated stage, takes hours to weeks per ion, and has scarcely changed in 50 years. It is exactly where modern techniques — regularization, Bayesian inference with priors, MCMC for *per-line gf uncertainties*, global optimization, and (our novel idea) folding the observed *stellar spectrum* into the objective alongside the energy levels — can make a genuine advance. See the project plan.

3 Mapping to the actual code and files

Program	Source (for/)	Computes
RCN	RCN36K.F	Single-config. radial wavefns $P_{n\ell}(r)$; radial integrals $E_{av}, F^k, G^k, \zeta_{n\ell}$
RCN2	RCN2K.F	CI integrals R^k (+ overlaps, off-diagonal spin-orbit); multipole transition integrals $\langle r \rangle$
RCG	rcg11k.f	Angular coeffs (Racah/cfp); builds & diagonalizes $H \rightarrow$ levels, eigenvectors, <i>gf</i>
RCE	RCE20K.F	Least-squares fit of radial parameters to observed levels; improved eigenvectors $\rightarrow$ <i>gf</i>

Table 1: The four programs and what they compute. Built by `build/build.sh`.

Data flow between programs. Each program writes the input for the next, mostly through fixed-name files (a legacy of the punched-card/tape era):

- A freshly compiled RCG must first generate, once, the binary “cfp decks” it needs: FOR072 holds the single-subshell quantum numbers and coefficients of fractional parentage; FOR073 and FOR074 hold derived matrix-element and angular-coefficient tables built from them. This is the `build/make_cfp.sh` step (it feeds the shipped `ING11.CFP` deck through RCG).
- In a calculation, RCG writes, among others, `TAPE2E` (binary coefficient matrices for RCE) and `OUTGINE` (a starter for RCE’s text input).
- RCE reads `TAPE2E` plus a text deck `INE20` (control info + the *observed* levels — the only thing not produced upstream) and writes `LEVELS1/2/3` (fitted levels and eigenvectors) and `PARVALS` (fitted parameter values, for restarting).

An actual input deck. The example `work/IN36` is an RCN input for Sn^{7+} (`IN36.ai` is B^{3+} , i.e. B IV, with a long Rydberg series). The first line is a fixed-column “control card” (mesh, method flags, convergence tolerances, relativistic/correlation switches); each subsequent line names one configuration. For Sn^{7+} :

```
200-90 0 2 01. 4.0 5.E-08 1.E-11-2 00090 0 1.0 0.65 0.0 1.00 -6
50 8Sn7+ 4d7 4p6 4D7
50 8Sn7+ 4d65p 4p6 4D6 5P
-1
```

Here the two configurations $4d^7$ and $4d^65p$ (opposite parity — the source and upper levels of E1 transitions) are listed, terminated by -1. The columns of this card are documented in `cowan_nist/extracted/RCN_DOC`. we will annotate them in detail when we drive a real run.

4 A worked example: Sn^{7+} end to end

For a single ion and one parity pair, the canonical chain is

$$\text{RCN} \rightarrow \text{RCN2} \rightarrow \text{RCG} (\rightarrow \text{RCE} \rightarrow \text{RCG}),$$

where the optional final RCE/RCG steps apply the semi-empirical fit. We have run the first three stages on the shipped example (`work/IN36+IN2`): Sn^{7+} with the even configuration $4d^7$ and the odd configuration $4d^65p$. The exact run lives in `work/sn7plus/`; the figures in this document are generated from its output by `docs/make_worked_figs.py`.

- RCN converged the self-consistent field for all nine subshells of each configuration and wrote the radial integrals to `out36` (e.g. for $4d^7$ it gives $\zeta_{4d} = 3762 \text{ cm}^{-1}$, $F^2(4d, 4d)$, $F^4(4d, 4d)$). Its converged orbitals are Fig. 1.
- RCN2 formed the CI and transition integrals and wrote `ING11`.
- RCG built and diagonalized the energy matrix for each J , producing ~ 1700 electric-dipole transitions between the two configurations, each with a wavelength, $\log gf$, and radiative rate. The resulting computed spectrum is Fig. 4 — a genuine (purely *ab initio*, not yet fitted) line list produced by our build of the engine.

The seven coupling schemes RCG/RCE support (LS, JJ, intermediate, ...) only change the basis, not the physics; final eigenvectors are reported in LS or JJ for interpretation.

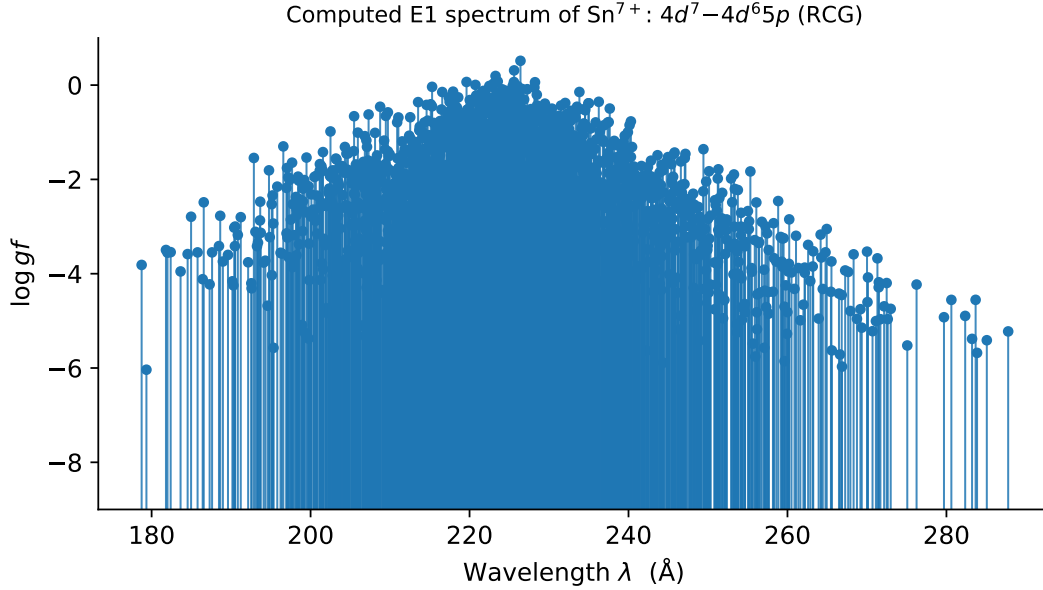


Figure 4: The computed electric-dipole spectrum of Sn^{7+} ($4d^7-4d^65p$): $\log gf$ versus wavelength for the ~ 1700 transitions from RCG. The band near 200–280 Å and the spread of line strengths over many orders of magnitude are typical of an open- d -shell transition array. These are raw *ab initio* values; the RCE fit (§2.4) would shift the levels onto observed energies and adjust the gf accordingly.

Next steps in the project.

1. Add observed levels and run RCE on this example to see the semi-empirical fit in action (replace the schematic Fig. 3 with the real computed-vs-observed level comparison).
2. Reproduce a known Kurucz Fe II gf via the standard chain (validate we replicate the legacy result).
3. Replace the RCE fit (7) with a modern regularized / Bayesian fit, attach per-line gf uncertainties, and (Tier 2) fold in the solar spectrum.

References

- R. D. Cowan, *The Theory of Atomic Structure and Spectra* (Univ. California Press, 1981). [TASS]
- Program documentation shipped with the code: RCN_DOC.txt, RCG_DOC.txt, RCE_DOC.txt (in `cowan_nist/extracted/`).
- A. Kramida, “Cowan Code: 50 Years of Growing Impact on Atomic Physics,” *Atoms* **7**, 64 (2019).